

Optimization for Machine Learning

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Review of Previous Class

- What is machine learning?
- Difference between software and AI model
- Applications of machine learning
- The role of machine learning
- Machine learning terms

Why Optimization Matters in Machine Learning?

- The main process of machine learning:

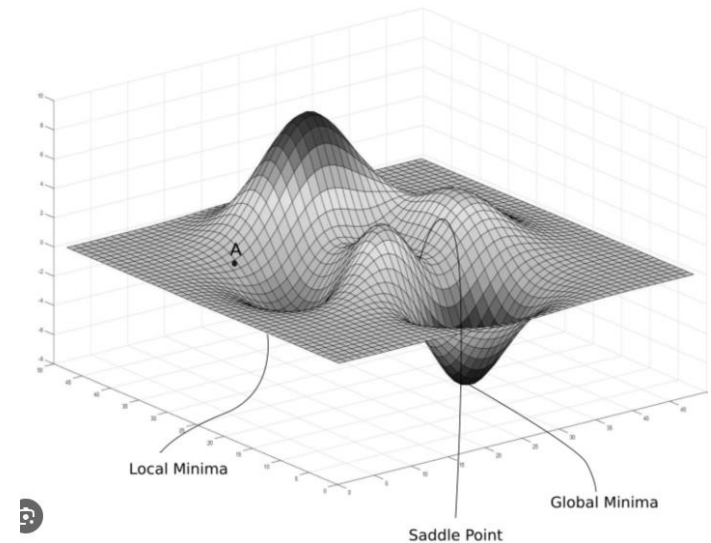
Data Collection → Model → Prediction → Loss →
Objective Function → Optimization → Learned Parameters

- The learned parameters are obtained through **optimization**.
- **Training a machine learning model** is often formulated as an optimization problem.

Convexity

Why Convexity?

- Optimization problems can be **very difficult** to solve
- Non-convex objective
- Multiple local minima
- Saddle points
- Difficult optimization landscape



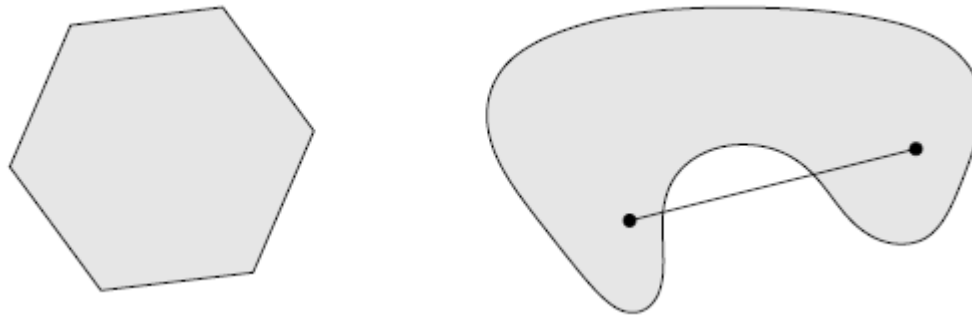
Convexity turns a difficult optimization problem into a more structured one.

Convex Set

- **Convex set:** $C \in \mathcal{R}^n$ is convex if:

$$\forall t \in [0, 1], x, y \in C \Rightarrow tx + (1 - t)y \in C$$

- Example:



- **Convex combination** of $x_1, x_2, \dots, x_k \in \mathcal{R}^n$: any linear combination:

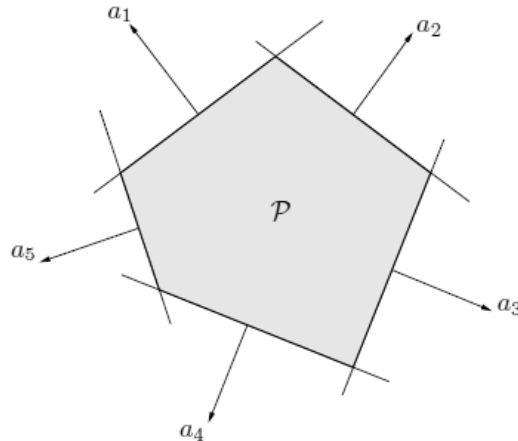
$$\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$

with $\forall i \in \{1, 2, \dots, k\}, \theta_i \geq 0$, and $\sum_{i=1}^k \theta_i = 1$.

- **Convex hull** $\text{conv}(C)$: all convex combinations of elements.

Examples of Convex Sets

- Empty set, point, line
- Norm ball: for given norm $\|\cdot\|$, radius r , $\{x : \|x\| \leq r\}$
- **Hyperplane**: for given a, b , $\{x : a^\top x = b\}$
- Halfspace: $\{x : a^\top x \leq b\}$
- Affine space: for given A, b , $\{x : A^\top x = b\}$
- **Polyhedron**: $\{x : A^\top x \leq b\}$



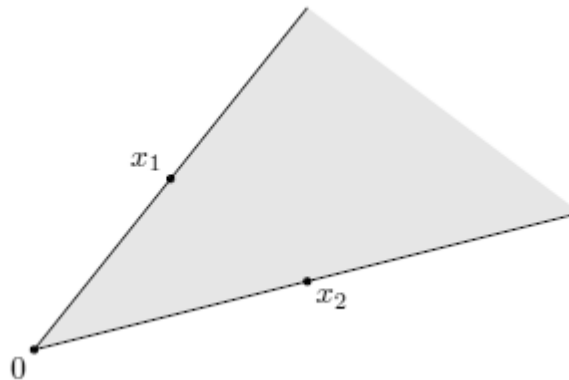
Cones

- **Cone:** $C \in \mathcal{R}^n$ is cone if:

$$\forall t \geq 0, x \in C \Rightarrow tx \in C$$

- **Convex cone:** cone that is also convex, i.e.,

$$\forall t_1, t_2 \geq 0, x_1, x_2 \in C \Rightarrow t_1x_1 + t_2x_2 \in C$$



- Conic combination:

$$x_1, \dots, x_k \in \mathcal{R}^n, \theta_1x_1 + \dots + \theta_kx_k$$

with $\forall i \in \{1, 2, \dots, k\}, \theta_i \geq 0$.

Conic hull collects all conic combinations.

Operations Preserving Convexity

- **Intersection:** the intersection of convex sets is convex
- **Scaling and translation:** if C is convex, then:

$$aC + b = \{ax + b : x \in C\}$$

is convex for any a, b

- **Affine images and preimages:** if $f(x) = Ax + b$ and C is convex, then:

$$f(C) = \{f(x) : x \in C\}$$

is convex, and if D is convex then:

$$f^{-1}(D) = \{x : f(x) \in D\}$$

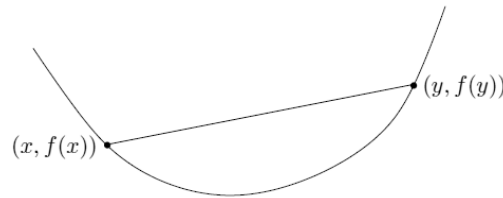
is convex.

Convex Functions

- **Convex function:** $f : \mathcal{R}^n \mapsto \mathcal{R}$ is a convex function if $\text{dom}(f) \subset \mathcal{R}^n$ is convex and:

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y), \text{ for } 0 \leq t \leq 1$$

and all $x, y \in \text{dom}(f)$.



- In words, functions lies below the line segment joining $f(x), f(y)$.
- **Concave function:** opposite inequality above, so that:

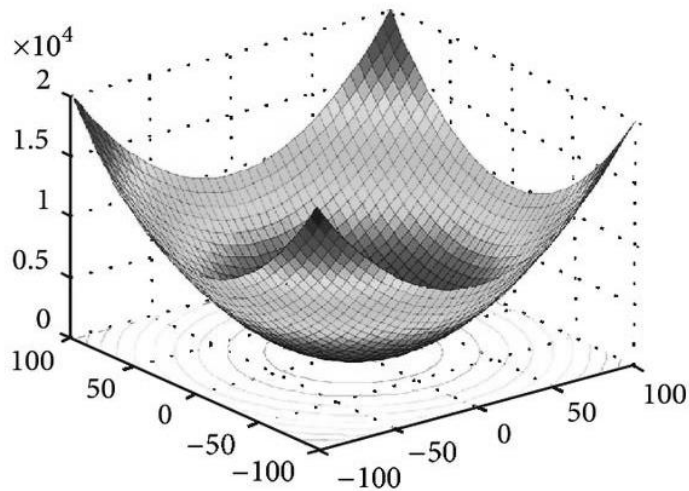
$$f \text{ is concave} \Leftrightarrow -f \text{ is convex}$$

Examples of Convex Functions

- Exponential function: $f(x) = e^{ax}$
- Power function: $f(x) = x^a$ is convex for $a \geq 1$ or $a \leq 0$ over $\mathcal{R}_+$
- **Logarithmic function** is concave over $\mathcal{R}_{++}$
- Affine function: $a^\top x + b$ is both convex and concave
- **Quadratic function:** $\frac{1}{2}x^\top Qx + b^\top x + c$ is convex provided that $Q \succeq 0$
- Least squares loss: $\|y - Ax\|_2^2$ is always convex

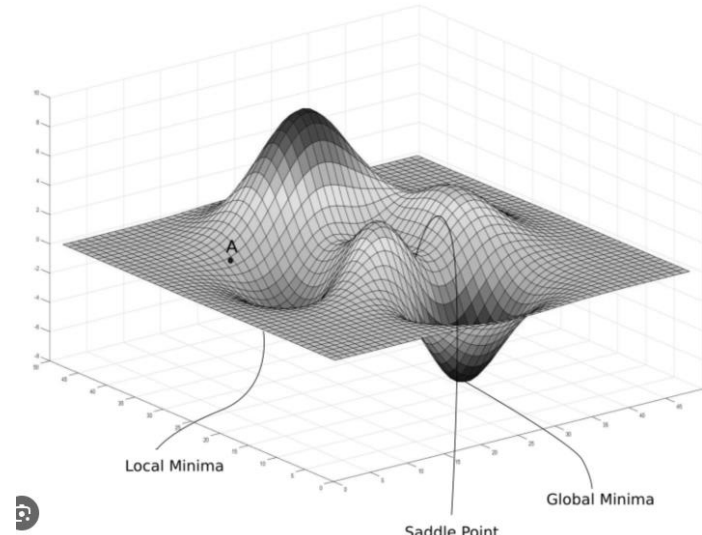
Why Convexity Matters?

- It gives us a much **simpler** optimization landscape.



Convex function

- Every local minimum is a global minimum
- Easier to optimize



Non-convex function

- Multiple local minima may exist
- Local minimum may not be global
- Optimization can be more challenging

Finding a local minimum is enough to find a global minimum.

Convex Optimization Problem

Optimization Problem

$$\begin{array}{ll}\min & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

- Objective function: $f_0(x)$
- Constraints: $f_i(x)$ and $h_i(x)$
- Optimal value p^* is defined as:

$$p^* = \inf\{f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, h_i(x) = 0, i = 1, \dots, p\}$$

- If x^* is feasible and $f_0(x^*) = p^*$, x^* is an **optimal point**.
- If there is an $R > 0$ such that:

$$\begin{aligned} f_0(x) = \inf\{f_0(z) \mid f_i(z) \leq 0, i = 1, \dots, m, \\ h_i(z) = 0, i = 1, \dots, p, \|z - x\|_2 \leq R\} \end{aligned}$$

x^* is **locally optimal point**.

Examples of Optimization Problem

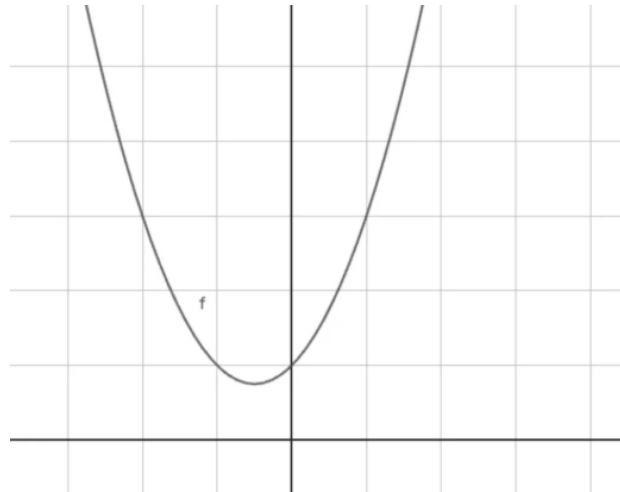
- Least-Norm Problem:

$$\min \|Ax - b\|_2$$

- Least-Norm Squared Problem:

$$\min \|Ax - b\|_2^2 = (Ax - b)^\top (Ax - b)$$

- These two problems are equivalent: the optimal points are the same



Convex Optimization Problem

$$\begin{aligned} \min \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & a_i^\top x = b_i \quad i = 1, \dots, p \end{aligned}$$

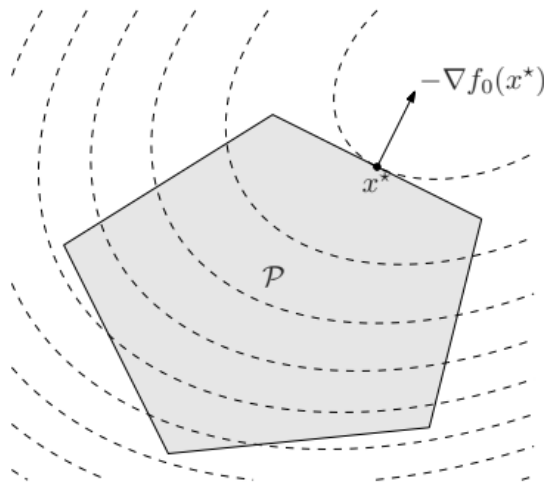
where $f_0, \dots, f_m$ are convex functions.

- The objective function must be convex
- The inequality constraint functions must be convex
- The equality constraint function $h_i(x) = a_i^\top x - b_i$ must be affine
- Any **locally optimal point** of a convex optimization problem is also **optimal**

Examples: Quadratic Optimization Problems

$$\begin{aligned} \min \quad & \frac{1}{2}x^\top Px + q^\top x + r \\ \text{subject to} \quad & Gx \preceq h \\ & Ax = b \end{aligned}$$

where $P \in \mathbf{S}_+^n, G \in \mathcal{R}^{m \times n}, A \in \mathcal{R}^{p \times n}$



Stochastic Gradient Descent

Methods for Optimization the Objective

- Closed-form solution
- Grid search
- Random search
- Gradient descent
- Stochastic gradient descent (SGD)
- Minibatch SGD

Example: Linear Regression

- Learn a model to Project X to Y .

X				Y
1	1	1	1	8
2	1	1	1	10
1	2	1	1	11
1	1	2	1	7
1	1	1	2	12
2	2	1	1	13

- How to project? Solving the following problem:

$$XA = Y$$

- Closed-form Solutions:

$$A^* = (X^\top X)^{-1} X^\top Y$$

Optimal Point

A^*
2
3
-1
4

1	1	1	1
2	1	1	1
1	2	1	1
1	1	2	1
1	1	1	2
2	2	1	1
X			

8
10
11
7
12
13
$\hat{Y}$

8
10
11
7
12
13
Y

Example: Linear Regression

- Learn a model to Project X to Y .

X				Y
1	1	1	1	8
2	1	1	1	10
1	2	1	1	11
1	1	2	1	7
1	1	1	2	12
2	2	1	1	13

- Optimization problem:

$$\min L = \|\hat{Y} - Y\|_2^2 = \|AX - Y\|_2^2$$

- Params: $A \in \mathcal{R}^4$

Optimal Point

A^*
2
3
-1
4

1	1	1	1
2	1	1	1
1	2	1	1
1	1	2	1
1	1	1	2
2	2	1	1
X			

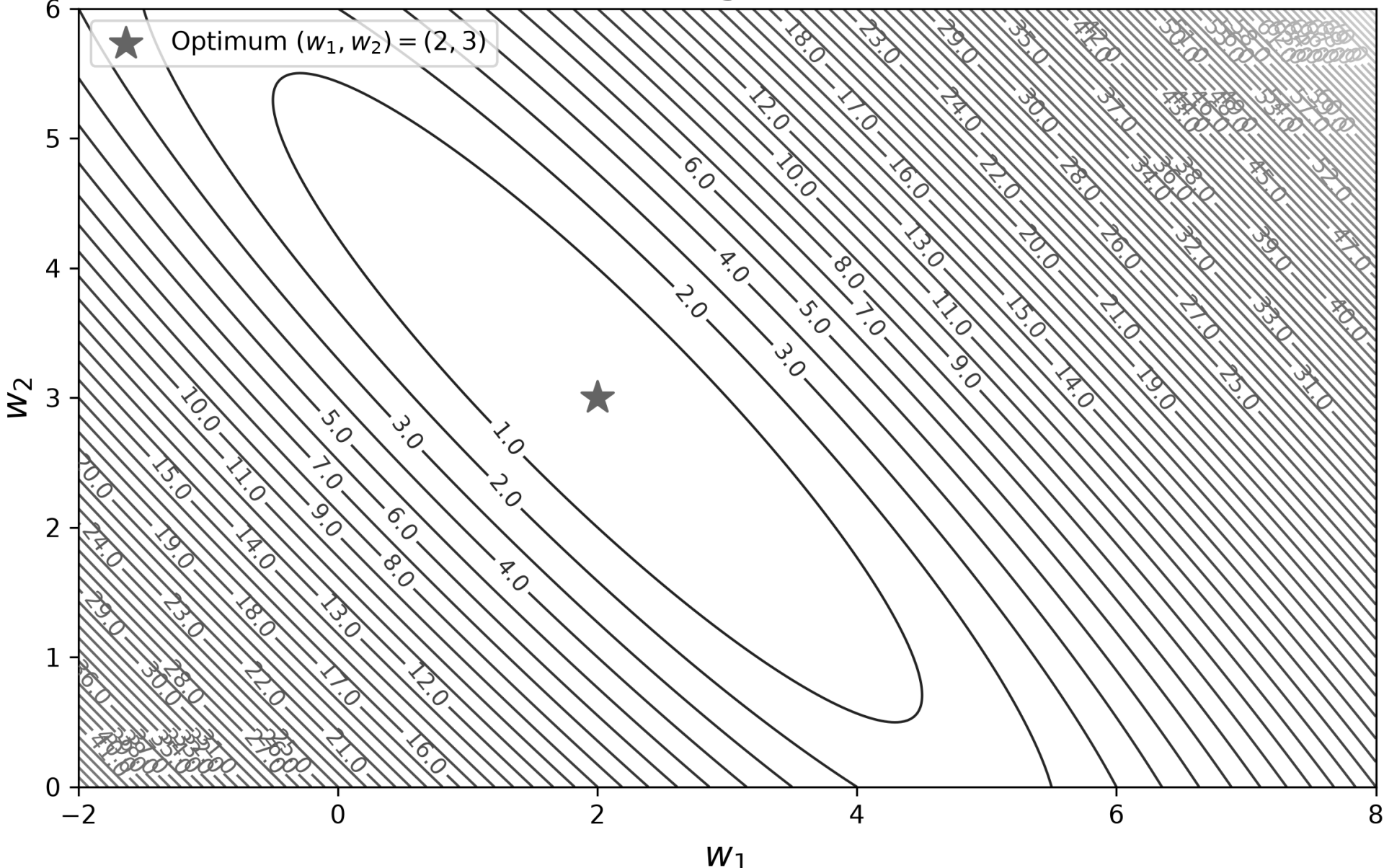
8
10
11
7
12
13
$\hat{Y}$

8
10
11
7
12
13
Y

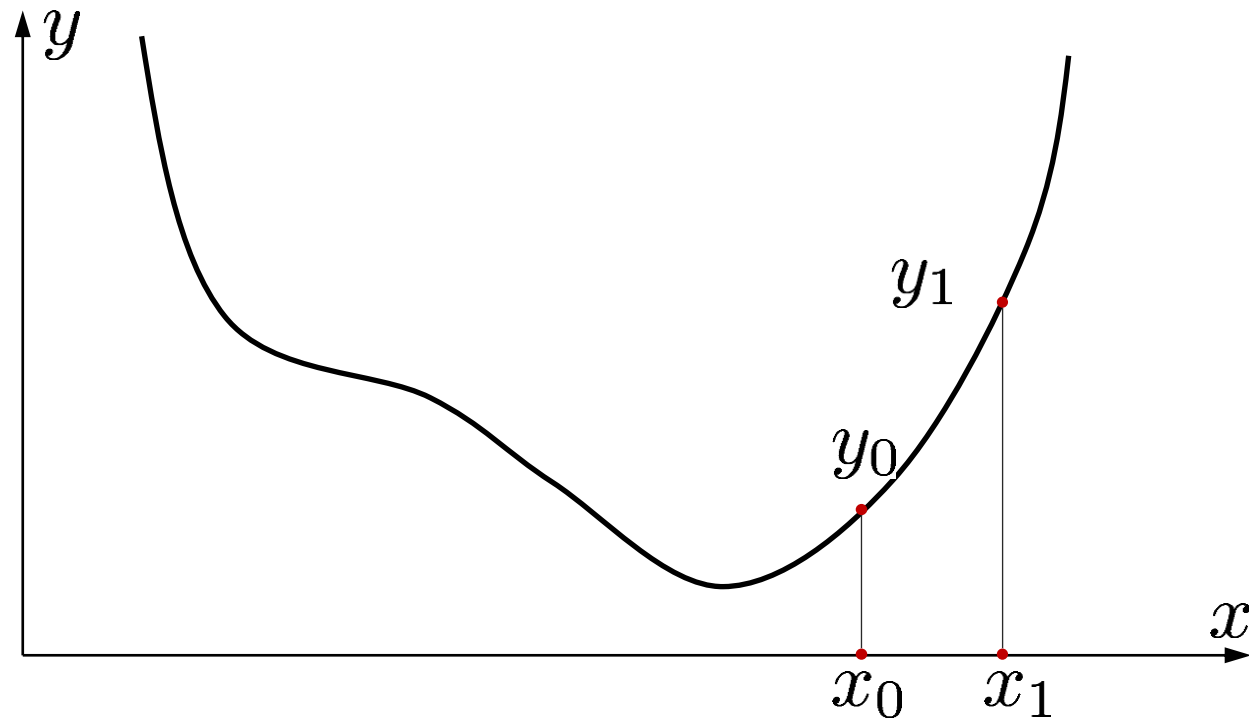
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Loss Contour Curve

Loss Contour of Linear Regression ($w_3 = -1, w_4 = 4$)

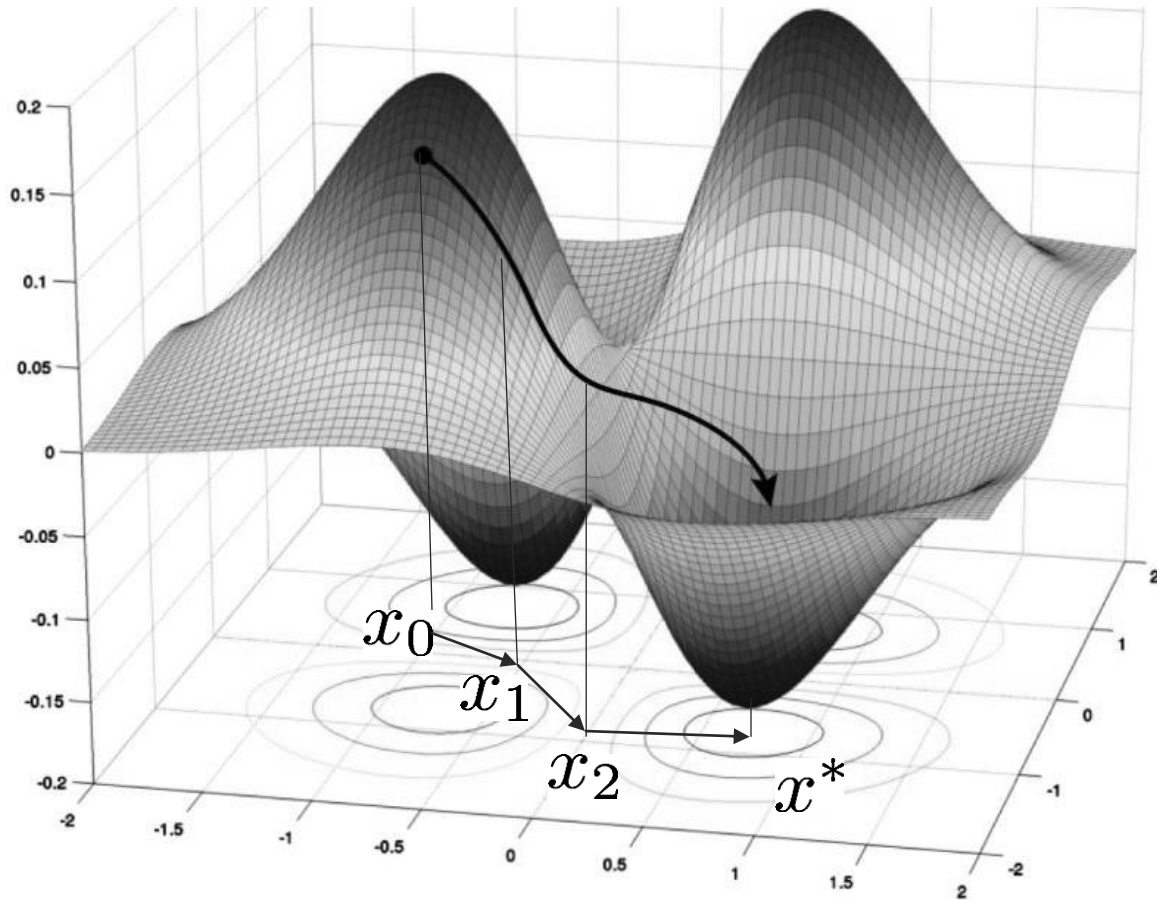


Gradient



The gradient points in the direction of the **steepest increase** of a function, while the negative gradient points in the direction of the **steepest decrease**.

Gradient



- The parameters x are updated **within the feasible set**.

Gradient

- Gradient of objective with respect to parameters:

$$\begin{aligned} L &= \|XA - Y\|_2^2 \\ &= (XA - Y)^\top (XA - Y) \\ &= \text{tr}(XAA^\top X^\top) - 2\text{tr}(XAY^\top) + \text{tr}(YY^\top) \end{aligned}$$

$$\frac{\partial L(X)}{\partial A} = 2X^\top XA - 2X^\top Y$$

- How to calculate the gradient?
 - Matrix Cookbook

Gradient

- Derivatives of Matrices, Vectors:

$$\frac{\partial x^\top a}{\partial x} = a$$

$$\frac{\partial a^\top X b}{\partial X} = a b^\top$$

$$\frac{\partial a^\top X^\top b}{\partial X} = b a^\top$$

$$\frac{\partial b^\top X^\top X c}{\partial X} = X(b c^\top + c b^\top)$$

$$\frac{\partial x^\top B x}{\partial x} = (B + B^\top)x$$

Gradient

- Derivatives of Trace:

$$\frac{\partial \text{tr}(X)}{\partial X} = I$$

$$\frac{\partial \text{tr}(XA)}{\partial X} = A^\top$$

$$\frac{\partial \text{tr}(AXB)}{\partial X} = A^\top B^\top$$

$$\frac{\partial \text{tr}(AX^\top B)}{\partial X} = BA$$

$$\frac{\partial \text{tr}(X^\top A)}{\partial X} = A$$

$$\frac{\partial \text{tr}(A^\top X)}{\partial X} = A$$

$$\frac{\partial \text{tr}(X^2)}{\partial X} = 2X^\top$$

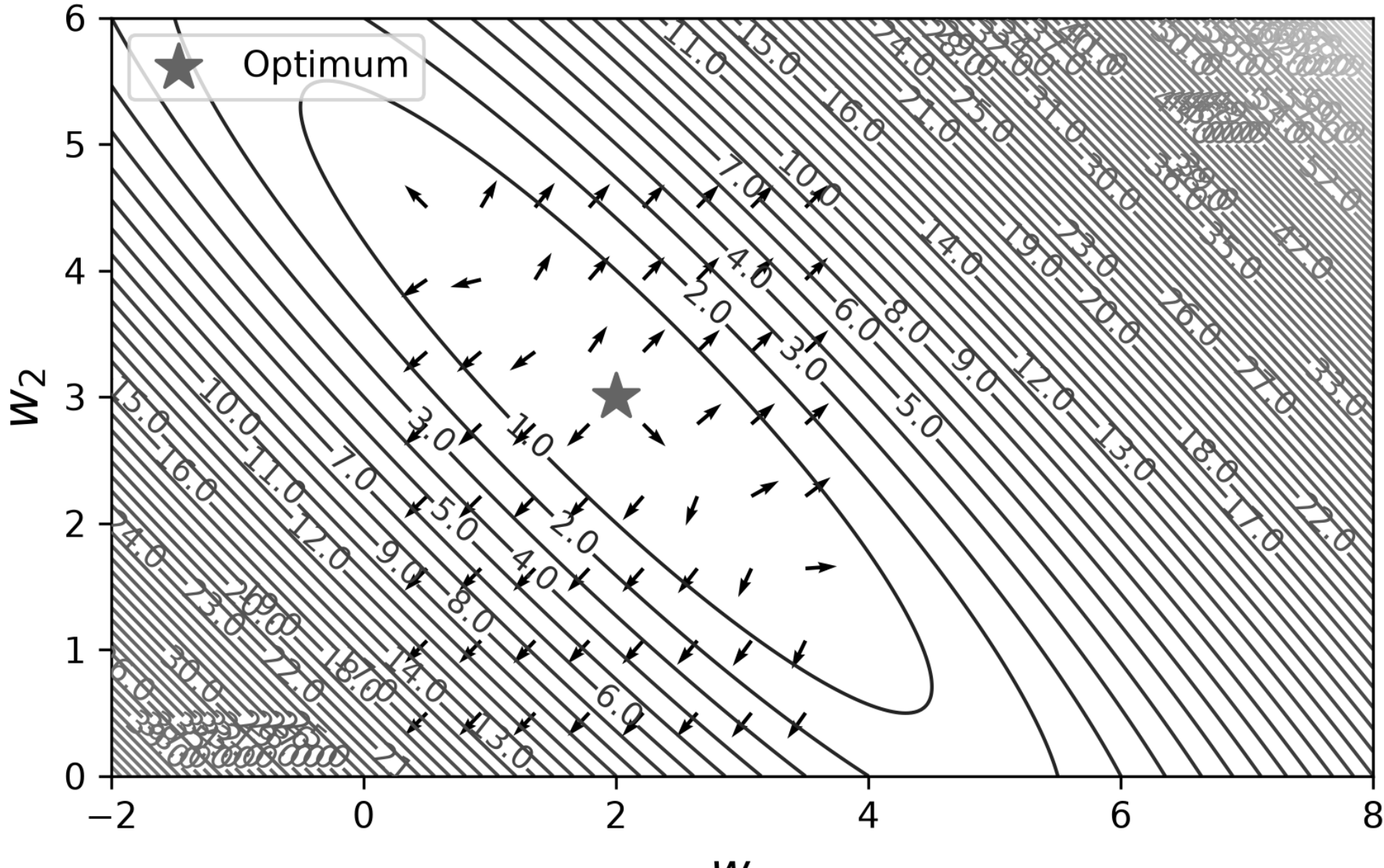
$$\frac{\partial \text{tr}(X^\top BX)}{\partial X} = (B + B^\top)X$$

$$\frac{\partial \text{tr}(BX^\top X)}{\partial X} = (B + B^\top)X$$

$$\frac{\partial \text{tr}(X^\top XB)}{\partial X} = (B + B^\top)X$$

Gradient

Gradient Direction



Gradient Descent

- How can we find the **minimum** of an objective function using its gradient? Gradient descent:

$$A_t = A_{t-1} - \eta \frac{\partial L}{\partial A}$$

- Iteration: t
- Learning rate: η
- Move step by step toward the optimum along the negative gradient direction.

Gradient Descent

- Gradient descent algorithm:

Algorithm 1 Gradient descent algorithm.

Input: Training set X and labels Y ;

Output: The learned models;

INIT: Initialize model parameter A_0 , learning rate η .

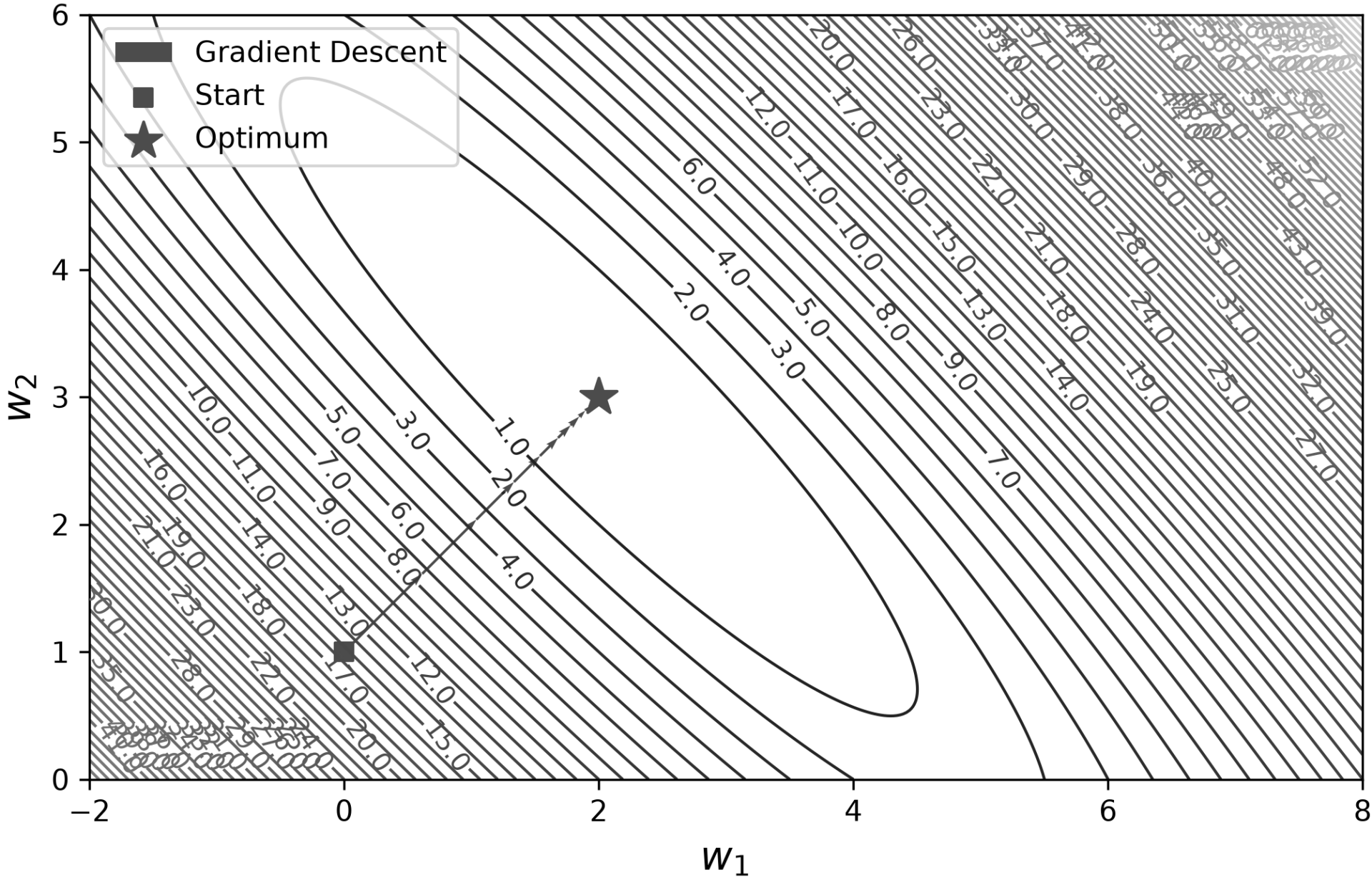
1: **for** $t = 1 \rightarrow \text{Num_Iteration}$ **do**

2: Calculate gradient $\frac{\partial L}{\partial A}$.

3: Update parameters: $A_t = A_{t-1} - \eta \frac{\partial L}{\partial A_{t-1}}$.

Gradient Descent

Gradient Descent



Stochastic Gradient Descent

- Gradient calculation requires all data points, which is time consuming:

Algorithm 1 Gradient descent algorithm.

Input: Training set X and labels Y ;

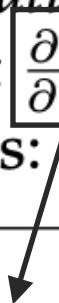
Output: The learned models;

INIT: Initialize model parameter A_0 , learning rate η .

1: **for** $t = 1 \rightarrow Num_Iteration$ **do**

2: Calculate gradient $\frac{\partial L}{\partial A}$.

3: Update parameters: $A_t = A_{t-1} - \eta \frac{\partial L}{\partial A_{t-1}}$.


$$\frac{\partial L(X)}{\partial A} = 2X^\top X A - 2X^\top Y$$

Stochastic Gradient Descent

- SGD: employ the gradient calculated by a **single data point** as estimation of full gradient.

$$\frac{\partial L(x_i)}{\partial A} = 2x_i^\top x_i A - 2x_i^\top Y$$

- Data point is sampled randomly.

Algorithm 2 Stochastic gradient descent algorithm.

Input: Training set X and labels Y ;

Output: The learned models;

INIT: Initialize model parameter A_0 , learning rate η .

1: **for** $t = 1 \rightarrow \text{Num_Iteration}$ **do**

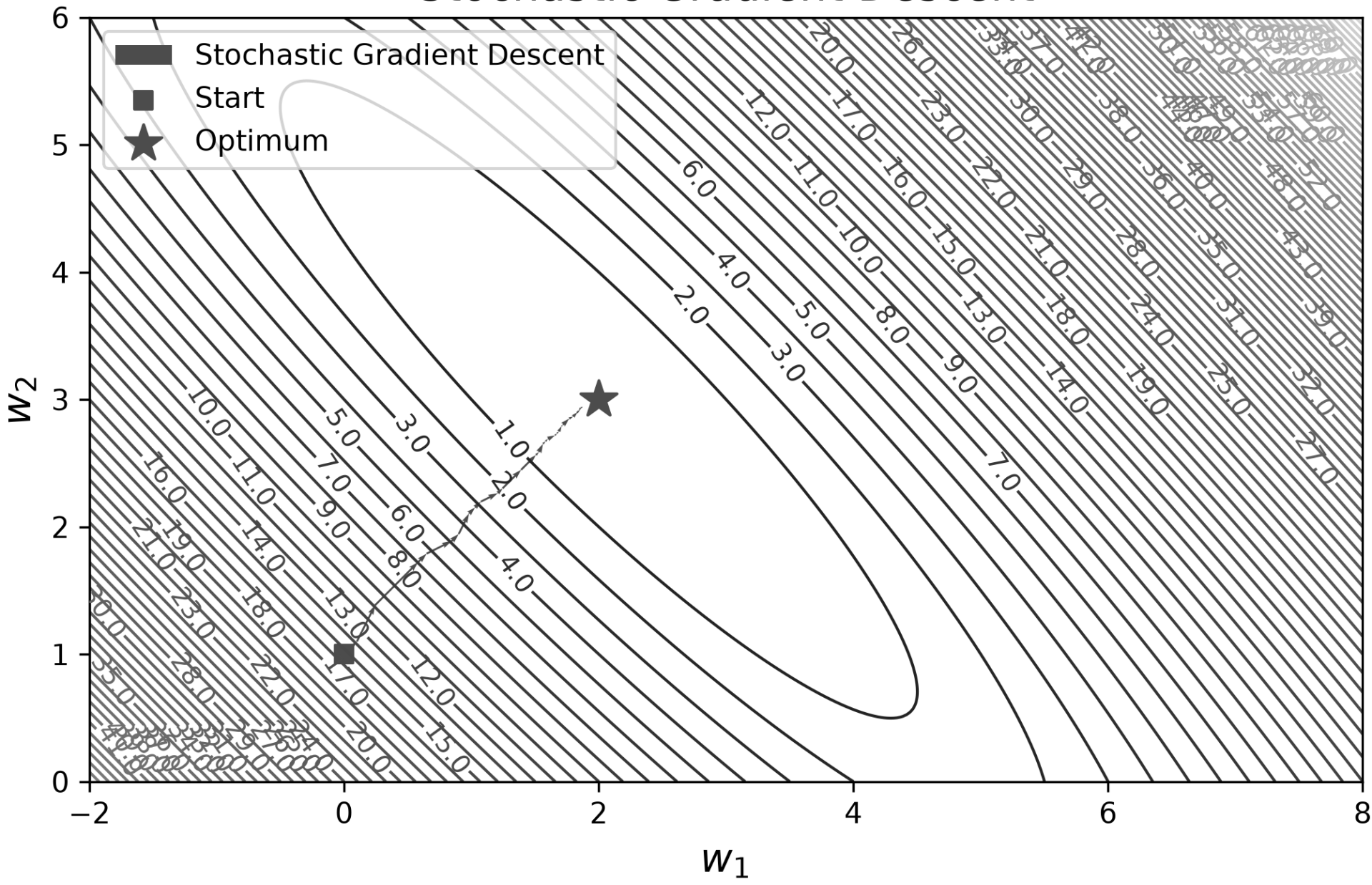
2: Sample a data point x_i randomly.

3: Calculate gradient $\frac{\partial L(x_i)}{\partial A}$.

4: Update parameters: $A_t = A_{t-1} - \eta \frac{\partial L(x_i)}{\partial A_{t-1}}$.

Stochastic Gradient Descent

Stochastic Gradient Descent



GD vs SGD

	Gradient Descent	Stochastic Gradient Descent
Gradient	Full dataset	One sample
Computation	Expensive	Cheap
Direction	Accurate	Noisy
Update frequency	Low	High
Memory	High	Low
Typical use	Small datasets	Large-scale learning

- Gradient descent: accurate but expensive
- SGD: noisy but efficient

What if we use a small batch of sample?

Why does SGD work?

- The full gradient is:

$$\frac{\partial L(X)}{\partial A} = \frac{1}{N} \sum_{i=1}^N \frac{\partial L(x_i)}{\partial A}$$

- SGD randomly samples a data point and uses:

$$g_i = \frac{\partial L(x_i)}{\partial A}$$

- If the data point is sampled uniformly, we have:

$$\mathbb{E}_{\xi: x_i \sim X} \left[\frac{\partial L(x_i)}{\partial A} \right] = \frac{\partial L(X)}{\partial A}$$

- Therefore:

The stochastic gradient is an **unbiased** estimator of the full gradient.

Mini-batch SGD: Accuracy or Efficiency?

- Mini-batch SGD: Accuracy or Efficiency?
- Instead of using **one sample (SGD)** or **the entire dataset (GD)**, we use a small batch of samples:

$$g_B = \frac{1}{B} \sum_{i \in \mathcal{B}_i} \frac{\partial L(x_i)}{\partial \theta}$$

Small batch	Large batch
More stochastic noise	More accurate gradient estimation
Cheaper update	More computation per update
More frequent updates	Less stochasticity

Mini-batch SGD provides a practical trade-off between gradient quality and computational efficiency.

Summary

- Learning = optimization of model parameters
- **Convexity** provides useful structure for optimization
- **Gradient** gives the steepest descent direction
- SGD improves scalability through stochastic estimation
- Mini-batch SGD balances efficiency and gradient quality

Optimization is the engine that turns a model into a learned model.

Thanks